Net force first

antocciphysics | TEACHER KEY | CPA

Learning target: Combine forces before using mass to predict acceleration.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 1, Day 10; 3 course days later.*
Earth pulls the sign down with 50 N; the cord pulls it up with 50 N.

02. Explain a worked example (worked)
Net arrow 6 N right; 9 N omits the opposing force.

03. Read the arrows (model)
12-4=8 N right; 8/4=2 m/s² right.

04. Balanced but moving (practice)
Net 0 N; a 0; keeps same leftward velocity.

05. Compare two masses (reason)
3 m/s² and 2 m/s²; 2 kg cart.

06. Choose a method (mixed) *Review: Unit 1, Day 1; 12 course days later.*
68 m if the 7 m/s rightward motion continues. Check a later timed position against the prediction.

07. Exit check (exit)
32 N right; 4 m/s² right.

08. STRETCH: Apply it another way (stretch)
Examples 8 and 2 N; 11 and 5 N. Both give 3 m/s² right.

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01. Bring back an earlier idea (retrieve) *Review: Unit 1, Day 10; 3 course days later.*
Weight 39.2 N down from Earth and tension 39.2 N up from string. Object-on-string acts on the string, not the object.

02. Explain a worked example (worked)
Net arrow 6 N right; 9 N omits the opposing force.

03. Work from force arrows (model)
Net 20 N right; a 4 m/s² right; v=-6+4=-2 m/s(left), slowing.

04. Find an unknown force (practice)
4-L=3(-2); L=10 N.

05. From braking to force (reason)
Forward positive; a=-6 m/s²; F=-9000 N.

06. Choose a method (mixed) *Review: Unit 1, Day 1; 12 course days later.*
v=-3.5 m/s; x(9)=-10.5 m. Assumes constant velocity; compare another timed position with its uncertainty.

07. Exit check (exit)
Net -12 N; a=-3 m/s². Need current velocity; leftward motion speeds up, rightward motion slows.

08. STRETCH: Apply it another way (stretch)
Net -6 N means other force -16 N. It can be moving right and slowing.